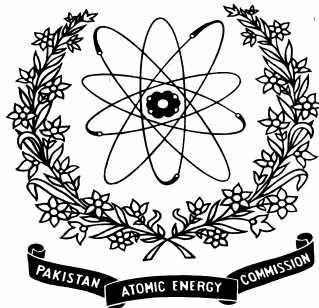


Radiation Dose Reduction by Water Shield

(W-Shielder Manual)



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ABSTRACT

This report is an operational manual of shielding software “W-Shielder”, developed at Health Physics Division (HPD), Pakistan Institute of Nuclear Science & Technology (PINSTECH), Pakistan Atomic Energy Commission. The software estimates shielding thickness for photons having their energy in the range 0.5 to 10 MeV. To compute the shield thickness, self absorption in the source has been neglected and the source has been assumed as a point source. Water is used as a shielding material in this software. The software is helpful in estimating the water thickness for safe handling, storage of gamma emitting radionuclide(s).

1. INTRODUCTION

Electromagnetic radiation possesses greater power of penetration than other nuclear radiation except neutrons.

When gamma radiation passes through matter, their intensity decreases due to interactions with the matter, such as photoelectric effect, Compton effect, pair production and photonuclear reactions. Use of matter to reduce the intensity of gamma radiation is known as shielding. The effectiveness of shield depends upon the energy of the radiation, thickness and type of the shielding material. The higher the atomic number and the density of the shielding material the more effective it is in reducing the intensity of gamma radiation.

In computing, the shield thickness, it has been assumed that the radiation source is a point source. It is possible to treat a line-source or other large radiation source as a point source if the distance from it amounts to more than twice the maximum of its extension. In this case, the highest theoretical deviation is about 4 % when computing the dose rate. [1-3].

This report discusses computer software “W-Shielder” developed in C++, which estimates the water thickness for shielding gamma radiation sources [4]. The software is helpful in designing shields and containers for safe handling and storage of gamma emitting radioactive materials such as spent fuel. The energy range of the photons considered is 0.5 to 10 MeV.

2. RADIATION SHIELDING

Gamma radiation shielding is the absorption and attenuation of gamma energy in shielding material. Most materials absorb the energy of gamma rays to some extent. The extent of attenuation depends on the density and thickness of the shielding material. A useful measure of shielding property is the mass per unit area of material. Hence a thick layer of a lighter material will have the same effect as a thin layer of a denser material. Radiation Shielding is used to reduce the radiation exposure to personnel and equipment in the vicinity of radiation sources. It is called biological and thermal shielding, respectively.

3. INTERACTION OF γ RADIATION WITH MATTER

Although more than ten distinctive elementary processes have been identified by which gamma rays interact with matter, for the purposes of radiation shielding, there are only three that are of any significance. These are:

- (i) The Photoelectric Effect (σ_{pe})
- (ii) Compton Scattering (σ_C)
- (iii) Electron-Positron Pair Production (σ_{pp})

The principal interaction processes for γ radiation in the range 0.1-10 MeV that are responsible for their attenuation are briefly given below.

3.1 Photo Electric Effect.

In this effect the incident photons disappears and an orbital electron is ejected from the atom. The electron carries away all the energy of the photon minus the binding (ionization) energy of the electron in the atom. This is an important mode of interaction for low energy photons in materials with high atomic numbers. The process can be regarded effectively as purely absorptive, since neither the photoelectrons nor the fluorescent radiation released as a consequence of photoelectric effect have sufficient energy to warrant independent study in shield analysis. The photoelectric cross-section depends strongly on Z , varying as

$$\mu_{pe} \sim Z^n$$

where n is the function of E . Because of the strong dependence of μ_{pe} on Z , the photoelectric effect is of greatest importance for the heavier atom such as lead, especially at lower energies [5].

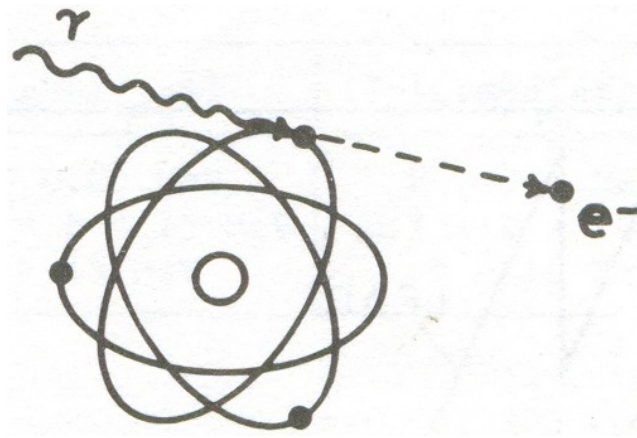


Fig. 1. Schematic diagram of the Photoelectric effect. The incident photon is 'destroyed'.

3.2 Compton Scattering

In this process a photon collides with an atomic electron loses some of its energy and is deflected from its original direction. It has an effect on the incident photon that is somewhat analogous to the effect that elastic scattering by an atomic nucleus has on a neutron – in both cases the incident particle survives the collisions. Compton scattering is the predominant reaction for gamma photons with energies in the range 1 MeV-10 MeV for elements of low and intermediate atomic numbers. The main theoretical difficulty in gamma attenuation calculation is attributable to the ramifications of multiple Compton scattering, it is, therefore, an important process in the theory of radiation shielding. The Compton cross-section per atom, σ_C , is equal to the number of electron in the atom-namely, Z -multiplied by σ_e . Thus,

$$\sigma_C = Z \sigma_e$$

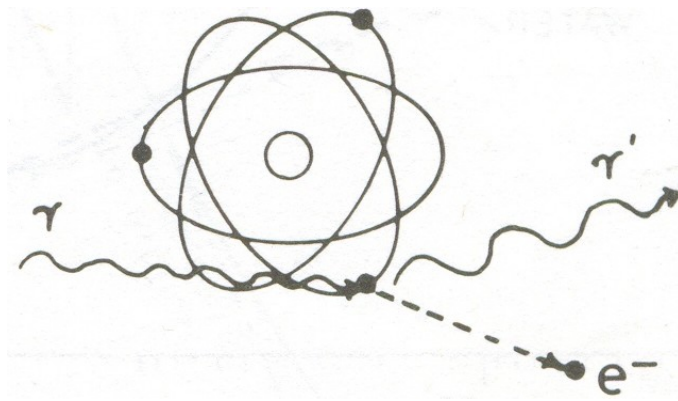


Fig. 2. Schematic diagram of the Compton effect. The incident photon is survives.

3.3 Pair Production

In this effect, which is the most likely interaction process for high energy photons, the incident photon disappear in the field of a charged particle, and an electron – positron pair appears. The cross section of this process varies as the square of the charge of the target particle – which may be an atomic electron, or, more usually, an atomic nucleus. In the former event the partial cross section is proportional to Z and in the later it is proportional to z^2 . Thus the total pair production cross-section increases rapidly with atomic number. To first order approximation the pair production event can be considered absorptive and the total energy of the incident photons assumed dissipated in the immediate vicinity of the interaction. Closer investigation reveals, however, that “annihilation gamma rays”, of relatively low energy, are generated as a consequence of the pair production process. In more exacting attenuation studies it is some times necessary to include explicitly these annihilation gamma rays.

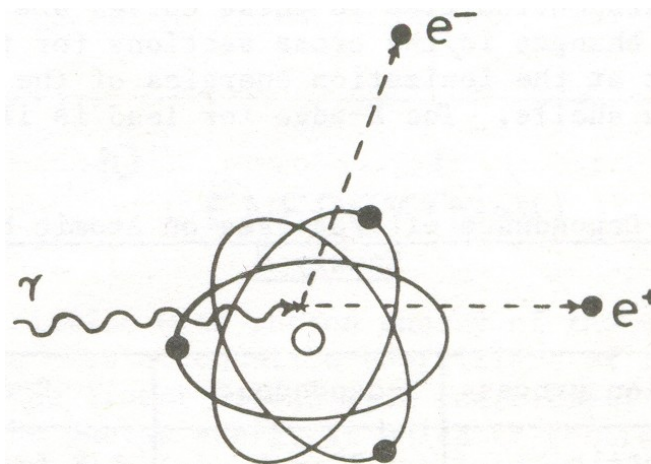


Fig. 3. Schematic diagram of the Pair production. This interaction can only occur if the incident photon has an energy exceeding 1.02 MeV. The incident photon is ‘destroyed’.

The total photon cross section per atom is obtained as the sum of these three partial cross sections, and we may write

$$\sigma = \sigma_{pe} + \sigma_C + \sigma_{PP}$$

and therefore, using an obvious notation,

$$\frac{\mu}{\rho} = \frac{\mu_{pe}}{\rho} + \frac{\mu_C}{\rho} + \frac{\mu_{PP}}{\rho}$$

and

$$\mu = \mu_{pe} + \mu_C + \mu_{PP}$$

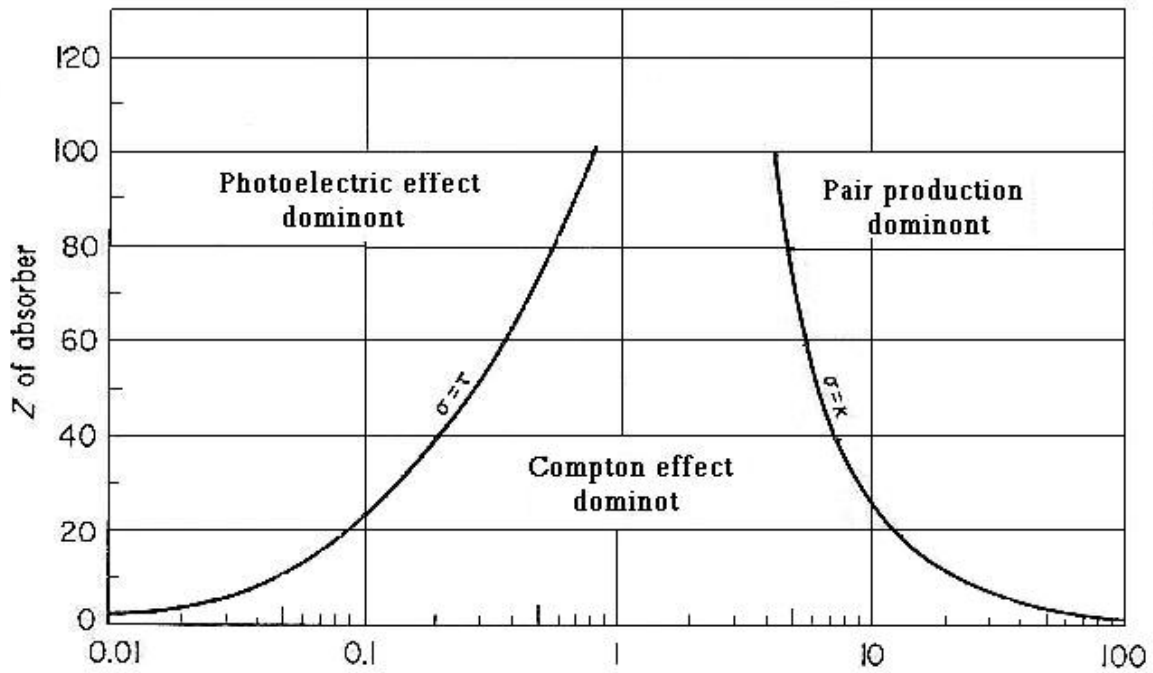


Fig. 4. Relative importance of the three major types of gamma-ray interaction.

Fig. 4 shows the relative importance for the three major types of gamma-ray interaction. The lines show the value of Z and $h\nu$ for which the two neighboring effects are just equal [6].

4. ATTENUATION OF γ RADIATION IN MATTER

4.1 Narrow Beam Geometry

In narrow beam geometry, a collimated beam of radiation is considered to be incident on a block of material containing **N atoms/cm³**. As a result of interaction the intensity **I** of γ ray reduces as the beam travels in the shielding medium. The reduction in the intensity, **dI** , in crossing a thickness **dx** in the shield is given by

$$dI = -IN\sigma_{\text{tot}}dx \dots\dots\dots(1)$$

The integrations gives;

$$I = I_0 \exp(-N\sigma_{\text{tot}}x) \dots\dots\dots(2)$$

Where **I_0** is the incident intensity and **I** is the residual intensity after passing a material of thickness “ **x** ”. **$N\sigma_{\text{tot}}$** is the macroscopic interaction cross-section, and is alternatively known as **μ** , the linear attenuation coefficient.

Equation (2) then becomes,

$$I = I_0 \exp(-\mu x) \dots\dots\dots(3)$$

In narrow beam geometry, the radiation source is placed at **S** , and beam is collimated by the collimator **A** . It then passes through an absorber **C** , through a collimator **B** and into the detector **D** . Under good collimating conditions, any interaction in **C** will appear as complete attenuation at the detector **D** , since scattered radiation cannot enter the detector because of the collimator **B** .

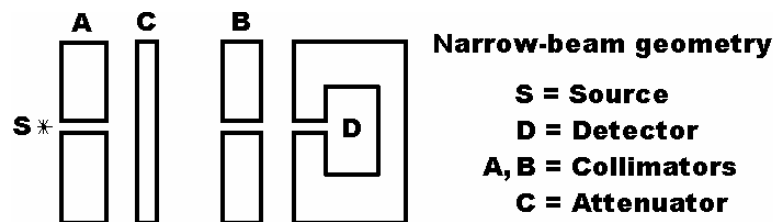


Fig.5. Narrow-beam geometry

Variations of linear attenuation coefficient μ for several materials are shown in Fig.6 and for water is given in Table 3. [7] The larger values of the attenuation coefficients of the heavier elements can clearly be seen.

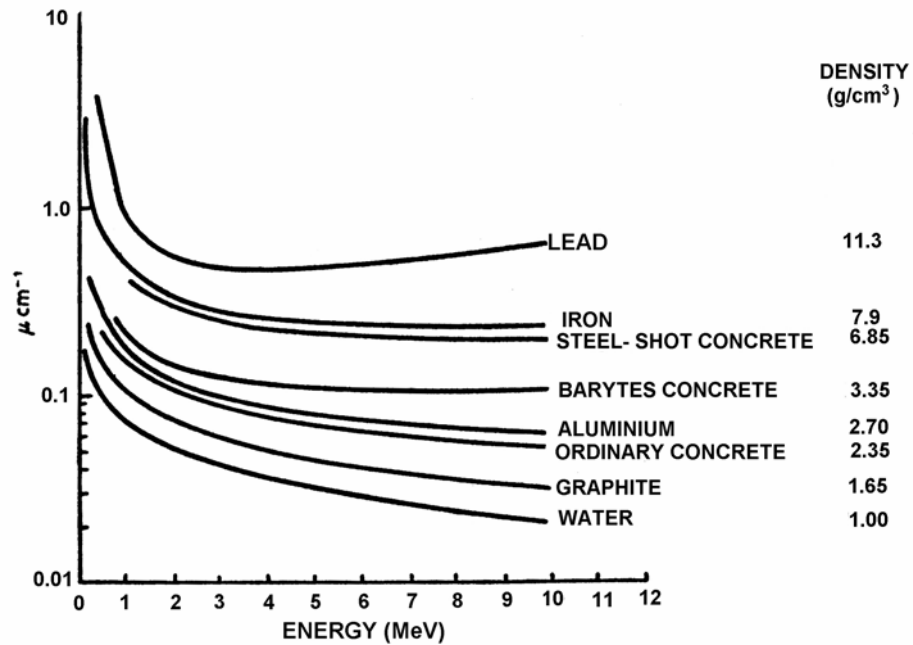


Fig. 6. Gamma-ray attenuation coefficients ($\mu \text{ cm}^{-1}$)

4.2 Broad Beam Attenuation

For broad and complicated source geometry beam attenuation, the dose at any point **P** after a shield will be greater than that predicted by narrow beam attenuation calculations, since these calculations take no account of scattered radiation.

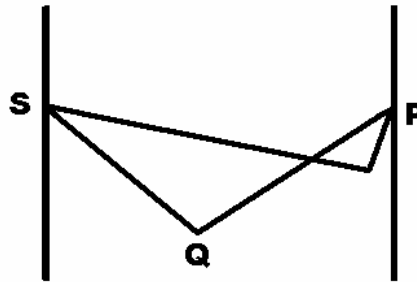


Fig. 7. Broad beam geometry.

The dose at **P** will be the sum of the uncollided flux ($I_0 \exp(-\mu x)$), at **P** and, radiation which undergoes scattering at **Q** and reaches **P**. Since the scattered radiation results from Compton processes, the effect will be more significant for the lighter elements where the Compton effect is the dominant process [7].

Allowance for this build-up of radiation intensity due to scattered radiation is made by the introduction of a build-up factor.

4.3 Homogeneous Media

Consideration of narrow beam experiment illustrated in Fig. 8.

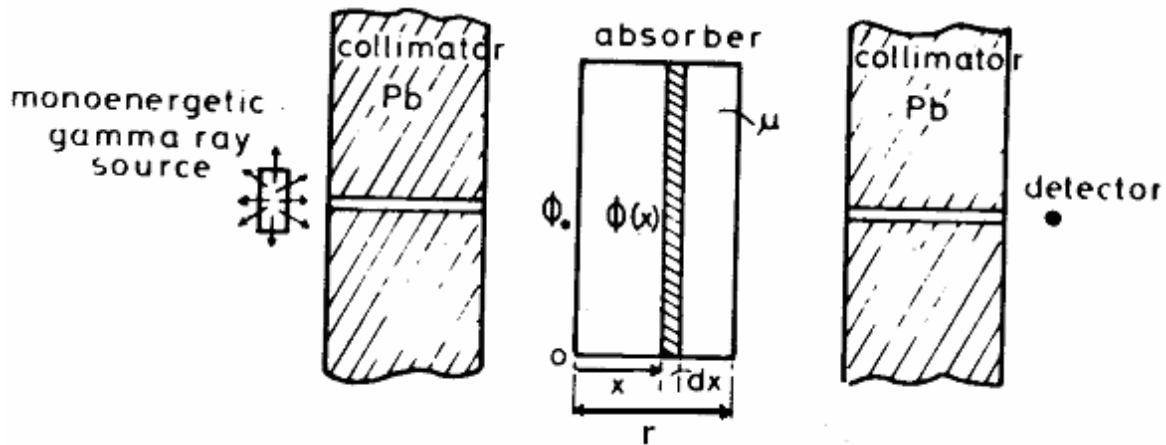


Fig. 8 Geometry of narrow beam experiment used to measure the linear attenuation, μ , of the absorbing slab.

Shows that if a detector is responsive only to those photons from the source that escape interaction in their passage through the intervening medium, then the detector response (to the 'uncollided' photon) can be expressed in the simple exponential form

$$\phi = \phi_0 e^{-\mu r} \dots\dots\dots(4)$$

However, in most shielding arrangements of practical importance, the gamma radiation reaching the point of interest (e.g the human body) will consist of, in addition to the uncollided component, a component consisting of :

- (i) photons that have undergone scattering interactions in the intervening medium, and
- (ii) low intensity, subsidiary radiation released locally in the intervening medium, due to the interaction between the source photons and the medium: annihilation gamma rays are an example of this type of radiation.

Therefore, in the shielding arrangements most frequently encountered in practice is the total radiation received that is of interest: such arrangements are often described as having 'broad beam geometry', to distinguish them from the well collimated (or 'good geometry') arrangement. A typical arrangement of source and shield having broad beam geometry is shown in Fig. 9.

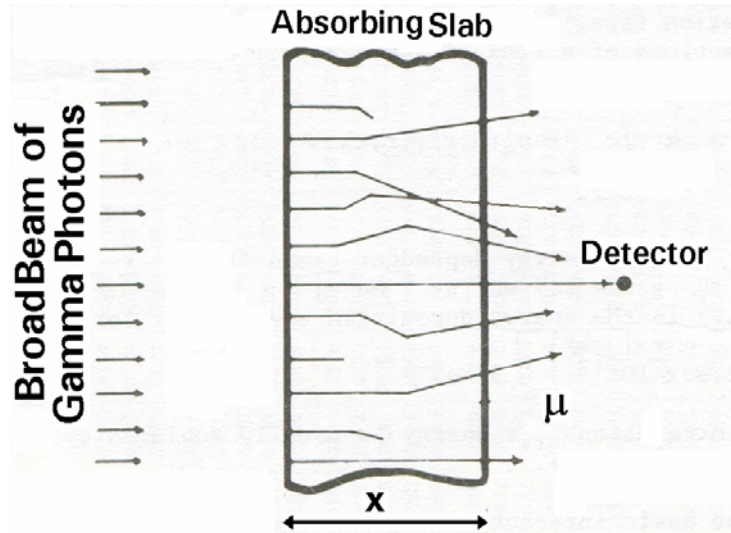


Fig. 9 Geometry of typical broad beam arrangement. In practice it is the total contribution of the radiation at the detector with which we are concerned; that is the uncollided component + the scattered component + locally released subsidiary radiation.

The variation of the flux Φ measured by the detector in Fig. 9, as a function of absorber thickness, is shown, qualitatively, Fig. 10.

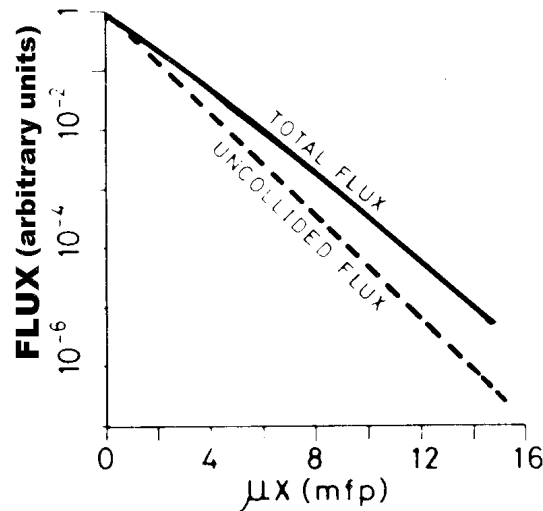


Fig. 10. Typical plot of flux variation with attenuating thickness for broad beam geometry. For comparison, the corresponding flux due to the uncollided photon is also shown.

4.4 Buildup Factor: the basic idea

In practical gamma ray shielding calculations the most widely used method of determining the total effect of the radiation at the point of interest makes use of a parameter called the buildup factor universally denoted by the symbol **B**. In principle

the buildup factor concept is also applicable to neutron attenuation, but in practice it is found to be much less successful when applied to neutrons, and consequently it is a much less important concept in the context of neutron shielding calculations. In the field of radiation shielding, unless stated otherwise, the term buildup factor can be assumed to refer to gamma radiation.

Briefly, the buildup factor can be considered as a simple correction factor which adjusts the result for the uncollided photons to include the contributions from the scattered and subsidiary radiation. The advantage of such a simple device is easily demonstrated. Let us take as a simple example, the calculation of the flux due to a monoenergetic unit point source of gamma rays at a point distance r in an absorbing medium. Here, for the sake of clarity in describing quantities of immediate concern, we use such terms as “point source” and “detector response” rather loosely, and rely on the reader’s intuitive understanding. The flux due to those photons that have penetrated the intervening material without suffering a collision, i.e. the unclouded flux, can be expressed immediately as

$$\phi_u = \frac{e^{-\mu r}}{4\pi r^2} \dots\dots\dots(5)$$

Now, assuming for the moment we know the values of the appropriate buildup factor B , we can readily write down the corresponding result for the total flux at r , as

$$\phi = B\phi_u = \frac{Be^{-\mu r}}{4\pi r^2} \dots\dots\dots(6)$$

In this way, we augment the effect of the uncollided photons to include the contributions from those photons that have undergone interactions in the surrounding medium, such as Compton scatterings, but have not been effectively ‘removed’ from the source radiation. Moreover, the formalism is such that what is intrinsically a formidable problem in particle transport theory, is reduced to the problem of obtaining the appropriate value of the buildup factor B . Note that in eq.(6), μ , is the usual narrow beam linear attenuation coefficient.

In practical terms, the value of B is obtained from the basic buildup data tables provided in shielding manuals. These basic data are mostly those of Goldstein and Wilkins [8] who used the moments method of solving the Boltzmann transport equation to derive extensive sets of buildup factor data for most materials of interest in shielding applications.

4.5 Buildup Factor: A more general approach

In shielding calculations there are other quantities of interest than the particle flux density, for example dose rate, and it is desirable to generalize the concept of buildup factor to include any quantity that can be measured by a detector. It is apparent from eq.(6) that B is the ratio of broad beam flux to the narrow beam flux.

We can utilize this relationship to extend the application of buildup factor, and define the buildup factor generally by;

$$B = \frac{\text{detector response to the radiation at the point of interest}}{\text{detector response to the uncollided radiation at the same point}}$$

The buildup factor is defined with respect to a monoenergetic-source and a detector situated in an infinite homogeneous medium.

Theoretical investigation shows that the value of **B** depends on the following conditions:

- (i) The nature of the attenuating medium (i.e. the atomic number, Z)
- (ii) The energy of the source photons.
- (iii) The distance, in mean free paths, between the source and point of interest.
- (iv) The geometry of the source.
- (v) The quantity being considered.

In view of this, the functional dependence of **B** can be expressed as

$$B(Z, E_0, \mu r)$$

where

Z	is the atomic number of the attenuating medium
E_0	is the energy of the monoenergetic source
μr	is the attenuating thickness of the intervening medium, in units of mean free path (mfp) of photons with energy E_0 .

Buildup factor data has been principally calculated for three detector responses, namely, exposure (often referred to as dose), energy deposition and particle flux. And for three particular source geometries: (a) point isotropic, (b) plane parallel and (c) plane isotropic. These three geometries are shown schematically in Fig. 11 (a, b, c).

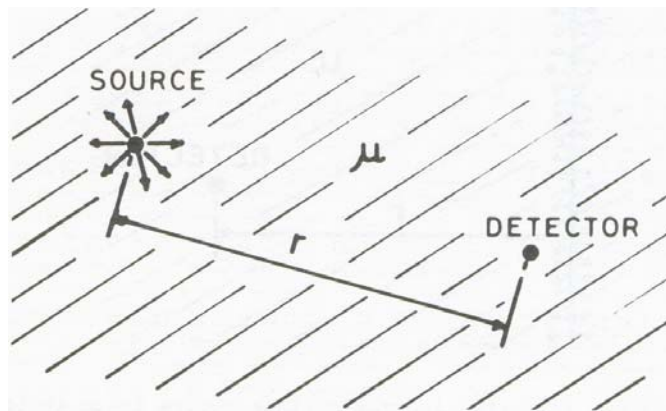


Fig. 11 (a) Point source in an infinite, absorbing medium.

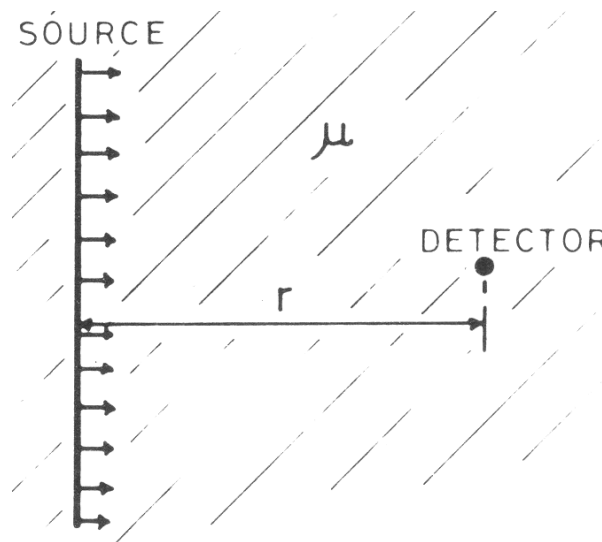


Fig. 11 (b) Infinite, normal plane parallel source in an infinite, absorbing medium.

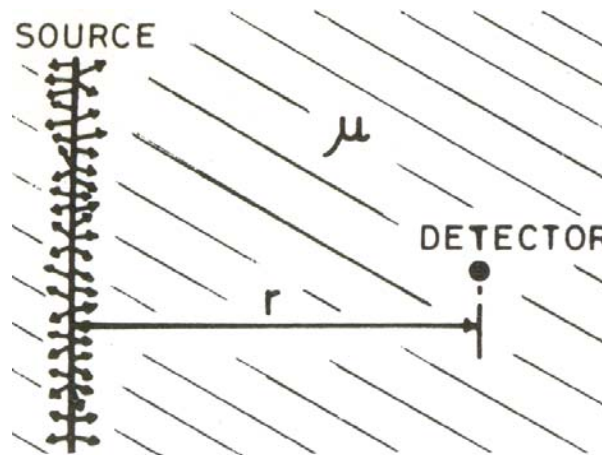


Fig. 11 (c) Infinite, isotropic plane source in an infinite, absorbing medium.

When obtaining buildup factor data from shielding manuals two points require special attention:

(i) **The nature of the quantity**

Usually the buildup factor is required for use in connection with the calculation of absorbed dose rate, in which case it is the tabulated values of exposure or dose buildup factor that should be used. On the other hand, in calculations of heat generated, it is the energy deposition buildup factor values that are more relevant.

(ii) **Source geometry**

Buildup factor values for a number of basic source geometries are usually provided; for example, for a point isotropic source.

Some typical values of buildup factors are shown in Table 1 and 2. These demonstrate the usual format employed: the user enters the table for a given material (i.e. atomic number Z), with the value of source energy E_0 , and the attenuating thickness μr in mfp's where μ is for photons of energy E_0 in the material of interest. In most shielding calculation, B roughly lies in the range 10 to 200. The dependence of B on Z and μr , for 1 and 8 MeV photons, is shown in Fig.12 [5].

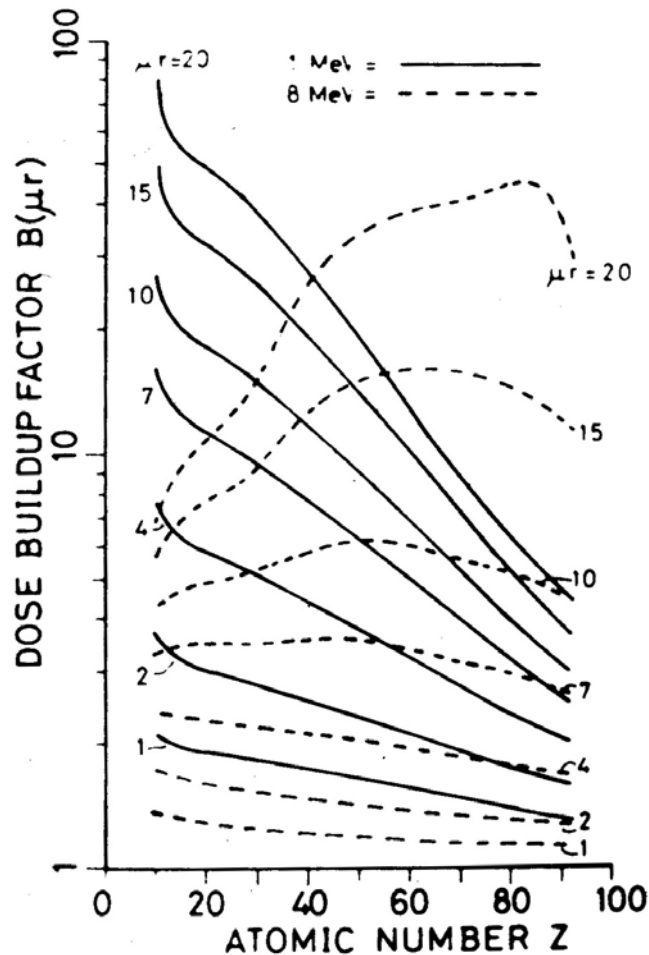


Fig. 12. Variation of buildup factor with atomic number

It is mentioned that most of the existing gamma ray buildup factor values, such as those shown in the tables 1 & 2, have been derived by treating only the Compton scattering accurately; the contribution from annihilation gamma rays, for example, have been neglected even for higher photons energies.

4.6 Empirical Formulae for $B(\mu r)$

In calculations involving the buildup factor it is convenient to have a mathematical expression for B , and a variety of such formulae have been proposed, the parameters of which have been adjusted to give a reasonable fit to the basic buildup factor data (mostly to original data of Goldstein [8] and viewed: some of the

more commonly used of these formulae are now briefly reviewed; a more extensive survey is given by Trubey [9]).

(i) Linear formula

$$B(\mu r) = 1 + k(\mu r)$$

if $\mu r = 1$ then the value of constant k will be

$$k = B(1) - 1$$

This is one of the simplest, and least accurate, approximations. It should not be used for thick shields. Its principal usefulness is didactic: for example in demonstrating the procedure for and effect of incorporating the buildup factor into shielding calculations.

(ii) Berger's formula

$$B(\mu r) = 1 + a\mu r \exp(b\mu r)$$

This formulation strikes a good balance between accuracy and computing complexity.

(iii) Capo's formula

$$B(E_o, \mu r) = \sum_{i=0}^3 \beta_i (\mu r)^i$$

Where

$$\beta_i = \sum_{j=0}^4 C_{ij} \left(\frac{1}{E_o}\right)^j$$

(iv) Taylor's formula

$$B = A \exp(-\alpha_1 \mu x) + (1 - A) \exp(-\alpha_2 \mu x)$$

The coefficient A , α_1 and α_2 in the preceding formulae is readily found in the various shielding manuals [10], and in Schaeffer [11]. Some values of the Taylor coefficients are given in Table 4. Caop's formula, although not particularly

convenient for mathematical working, has the advantage of being more general in as much as the source energy, E_0 , is included as a variable: the other formulae require a different set of coefficients, or parameters, for each value of E_0 .

Perhaps the most widely used is Taylor's formulation, as it lends itself to analytical integration in subsequent mathematical manipulation. Of course, the more complex the mathematical expression assumed for B , the better is the fit obtained to the original data over a wide range of μx . But if mathematical form is more complex than, say Taylor's, it makes subsequent analytical solutions difficult to drive.

5. THEORY BEHIND W-SHIELDER

This software employs **Taylor's** notation for build up factor, which is

$$B = A \exp(-\alpha_1 \mu x) + (1 - A) \exp(-\alpha_2 \mu x) \dots\dots\dots(7)$$

Where A , α_1 , α_2 are Taylor build-up coefficients and μ is the linear attenuation coefficient. These depend upon γ ray energy of radiation source and type of shielding material. Values of A , α_1 , α_2 and μ for water are given in Table 3 & 4.

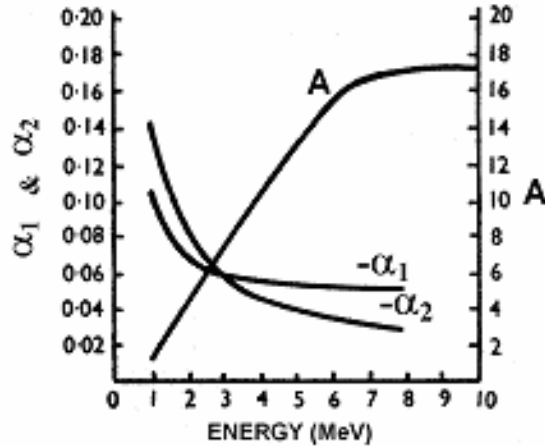


Fig. 13 Taylor build-up coefficients for water.

The intensity for collimated radiation source, including build-up, is provided by

$$\begin{aligned} I &= I_0 \exp(-\mu x) B \\ &= I_0 \exp(-\mu x) (A \exp(-\alpha_1 \mu x) + (1 - A) \exp(-\alpha_2 \mu x)) \\ &= I_0 A \exp-(1 + \alpha_1) \mu x + (1 - A) \exp-(1 + \alpha_2) \mu x \dots\dots\dots(8) \end{aligned}$$

This formulation replaces the incident γ ray attenuation coefficient μ by two slightly different attenuation coefficients $(1 + \alpha_1)\mu x$ and $(1 + \alpha_2)\mu x$, weighted by A and $(1 - A)$, respectively.

The expression obtained for the flux with build-up is thus very similar to that for the uncollided flux, $I_0 \exp(-\mu x)$, with slightly modified exponents and weighting factors.

6. HOW DOES THE SOFTWARE WORK

- By selecting any given radionuclide or entering energy in the “Main Menu”, software interpolates the value of μ , A , α_1 & α_2 for water against the given energy by using the parameters of Taylor build-up coefficients, as given in Table 3 & 4. These coefficients have already been fed in the software.
- Software takes the value of I_0 from “Activity-Doserate Menu” and “Reduced Doserate Menu” and then equates the eq. 9 to “1”.

$$1 = I_0 / I (A \exp-(1 + \alpha_1)\mu x + (1 - A) \exp-(1 + \alpha_2)\mu x) \dots(9)$$

- Software assumes the value of thickness “ x ” as “0.01” and calculates for it. If the R.H.S is equal to or very close to “1” then it quits the calculation.
- If R.H.S is not equal or close to “1” it increases the value of thickness “ x ” until R.H.S comes very close to “1”.
- Software draws a cylinder in accordance with calculated thickness.
- The software at the end displays a “Report” that shows the results of its interpolation and parameters for which the thickness was calculated.
- After pressing any key the software is ready to calculate shielding thickness again or exit from it.

7. HOW TO OPERATE THE SOFTWARE.

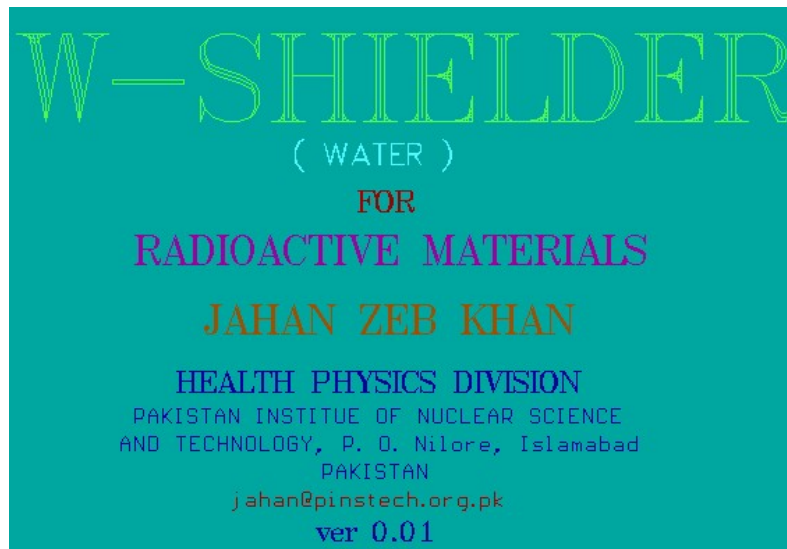
- Software consists of a single executable file “W-Shielder.exe” which can be run both in DOS and Window environment.

DOS environment:

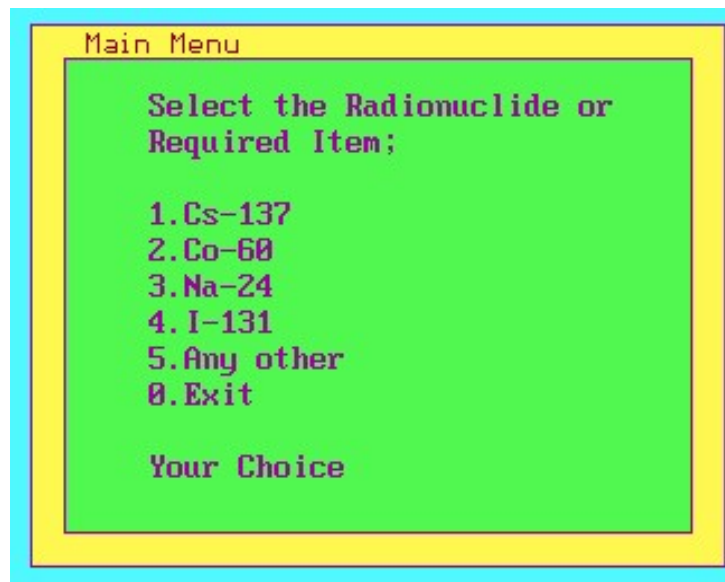
- Copy W-Shielder.exe at your desired location i.e. C:\W-Shielder or in any other folder.
- Log into desired location and then type “W-Shielder” or otherwise A:\W-Shielder

Window environment:

- Click “Start”, select “Run” and then type “A:\W-Shielder” and then select “OK”.
Title screen for W-Shielder will appear on the monitor and will disappear on pressing any key (Any key does not include all functions keys, Alt, Ctrl & Shift keys) and Main menu screen will appear on the monitor.
- The Main menu screen looks like:



- After three seconds and “pressing any key” the following “**Main Menu**” window will appear.



- Here the user has to select the **Radionuclide** by using num keys "1", "2", "3" or "4". "0" key is used to exit from the software.

- If the desired radionuclide is not present in the window then using num key “5”, following window will appear.

Any-Radionuclide Menu

Please enter the Energy of the radionuclide (from 0.5 to 10.0 MeV)

1

Please enter the Gamma Factor of the radionuclide(R.cm²/h.mCi)

1

- Here user has to type the **energy (from 0.5 to 10 MeV)** and gamma factor (Γ) of the radionuclide and press the return key. Gamma factor is used to calculate dose rate if “Activity” is fed in next window. To design container for multiple radiation sources, energy and gamma factor of that radionuclide is entered which has max. value of (Ci.MeV).
- On selecting “1 through 4” in main menu or gamma factor in the previous window, the following window will appear.

Activity-Doserate Menu

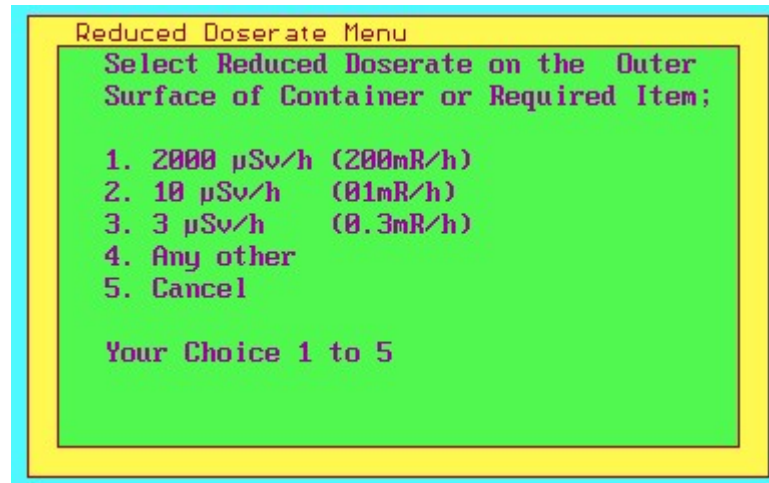
Select the Required Item;

1. Activity
2. Dose rate
3. Cancel

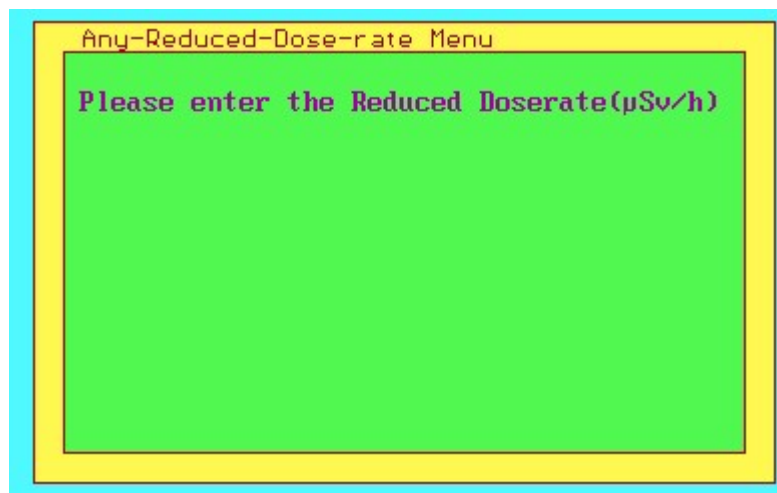
Your Choice 1 to 3

- Now user has to select the Activity or Dose rate depending upon the data he has.
- For selecting “**Activity**” use num key “1” and type value of Activity in “**mCi**” and press return key.

- For selecting “**Dose rate**” use num key “2” and type value of Dose rate in “ $\mu\text{Sv/hr}$ ” from the source at point of observation or without shielding and press return key.
- After typing Activity or Dose rate, the next window of different reduced Dose rates will appear.

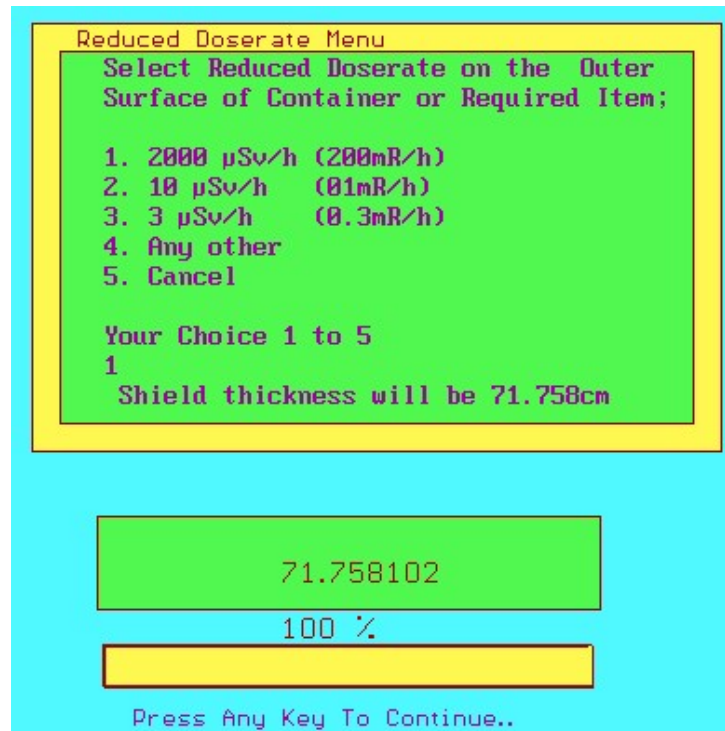


- Select the **reduced/required dose rate** on the outer surface of water.
- If user's desired dose rate is not present then use num key “4”, which represents “**Any other**” reduced dose rate and following window will appear.



- User has to enter the **required reduced dose rate** in “ $\mu\text{Sv/h}$ ”.

- The software will start calculating the shielding and show the results as below.



- After calculating, the software displays the shielding thickness and draws a cylinder in the next window with wall thickness as shield thickness.
- Gap between source surface and cylinder's inner surface is 1 cm.



- The software displays the details of its calculations in the **Report** window, which also includes the interpolated values of different parameters of Taylor build-up coefficients, and mass attenuation Coefficient of water.

Report	
Dose rate(mSv/h)	33.000
Reduced Dose rate(μ Sv/h)	2000.000
Energy(Mev)	0.662000
r (R.cm ² /h.mCi)	3.300000
A for Water	74.521942
- α 1 for Water	0.115044
α 2 for Water	-0.082024
Density of Water (g/cm ³)	1.000000
Mass att. Coeffcient (cm ² /g)	0.086190
Linear att. Coeffcient (1/cm)	0.086190
Shield thickness (cm)	71.758102
Shield thickness (inch)	28.251221
Press Any Key To Continue..	

- On pressing any key, main menu will appear again for the reuse or to exit the programme.

8. ATTENUATION FACTOR TABLE

The shielding thickness required to achieve a given attenuation factor is presented for water in table 6 by using the “W-Shielder” software. This table generated by “W-Shielder” software can be used to estimates thickness of water column requires to attenuate/reduce the dose rate by a factor up to 10^7 times due to gamma rays having energy in the range from 0.5 to 10 MeV.

Case – I

What will be the thickness of a water column to reduce a doserate of 10 Sv.h^{-1} to $1\mu\text{Sv.h}^{-1}$ due to a gamma ray of 5 MeV?

Solution:

$$D1 = 10 \text{ Sv.h}^{-1}$$

$$\begin{aligned}
 D2 &= 1\mu \text{ Sv.h}^{-1} \\
 \text{Attenuation Factor} &= D1/D2 = 10 \text{ Sv.h}^{-1}/1\mu \text{ Sv.h}^{-1} \\
 &= 10^7
 \end{aligned}$$

So, from the table 5, column 7 for 5 MeV and row 9 for attenuation factor 10^7 , the thickness will be **611.92 cm**.

Case – II

A Co-60 source (energy of gamma radiation $E = 1.25 \text{ MeV}$) with an activity of 775.2 Ci is to be shielded with a water column so that the dose rate after shielding is $10 \mu\text{Sv.h}^{-1}$. What will be thickness of water column?

Solution:

$$\text{Activity} = 775.2 \text{ Ci}$$

$$\Gamma = 12.9 \text{ R. cm}^2/\text{h. Ci}$$

$$\text{So dose rate will be} = 775.2 \text{ Ci} \times 12.9 \text{ R. cm}^2.\text{h}^{-1}.\text{Ci}^{-1}$$

$$= 10000 \text{ R.h}^{-1}$$

$$= 100 \text{ Sv.h}^{-1}$$

$$D1 = 100 \text{ Sv.h}^{-1}$$

$$D2 = 10 \mu \text{ Sv.h}^{-1}$$

$$\text{Attenuation Factor} = D1/D2 = 100 \text{ Sv.h}^{-1}/10\mu \text{ Sv.h}^{-1} = 10^7$$

Since there is no data is given for 1.25 MeV energy. So, the table will be linearly interpolated with sufficient accuracy to estimates the thickness.

For the interpolation;

$$E = 1.25$$

$$E1 = 1$$

$$E2 = 2$$

$$T1 = 292.75 (\text{Values of thickness for 1 MeV at } 10^7 \text{ attenuation factor})$$

$$T2 = 393.94 (\text{Values of thickness for 2 MeV at } 10^7 \text{ attenuation factor})$$

$$\begin{aligned}
 \text{Thickness will be} &= T1 + ((T2-T1)*(E-E1))/(E2-E1) \\
 &= 292.75 + ((393.94-292.75)*(1.25-1))/(2-1) \\
 &= \mathbf{318.05 \text{ cm}}
 \end{aligned}$$

The thickness can also be estimated from the Fig. 14 which shows the water column thickness for different gamma energies and attenuation factors.

Table 1: Exposure Buildup Factor for Isotropic Point Source in an Infinite Medium [5]

Material	E_0 , Mev	μr						
		1	2	4	7	10	15	20
Water	0.255	3.09	7.14	23.0	72.9	166	456	982
	0.5	2.52	5.14	14.3	38.8	77.6	178	334
	1.0	2.13	3.71	7.68	16.2	27.1	50.4	82.2
	2.0	1.83	2.77	4.88	8.46	12.4	19.5	27.7
	3.0	1.69	2.42	3.91	6.23	8.63	12.8	17.0
	4.0	1.58	2.17	3.34	5.13	6.94	9.97	12.9
	6.0	1.46	1.91	2.76	3.99	5.18	7.09	8.85
	8.0	1.38	1.74	2.40	3.34	4.25	5.66	6.95
	10.0	1.33	1.63	2.19	2.97	3.72	4.90	5.98
Aluminum	0.5	2.37	4.24	9.47	21.5	38.9	80.8	141
	1.0	2.02	3.31	6.57	13.1	21.2	37.9	58.5
	2.0	1.75	2.61	4.62	8.05	11.9	18.7	26.3
	3.0	1.64	2.32	3.78	6.14	8.65	13.0	17.7
	4.0	1.53	2.08	3.22	5.01	6.88	10.1	13.4
	6.0	1.42	1.85	2.70	4.06	5.49	7.97	10.4
	8.0	1.34	1.68	2.37	3.45	4.58	6.56	8.52
	10.0	1.28	1.55	2.12	3.01	3.96	5.63	7.32
Iron	0.5	1.98	3.09	5.98	11.7	19.2	35.4	55.6
	1.0	1.87	2.89	5.39	10.2	16.2	28.3	42.7
	2.0	1.76	2.43	4.13	7.25	10.9	17.6	25.1
	3.0	1.55	2.15	3.51	5.85	8.51	13.5	19.1
	4.0	1.45	1.94	3.03	4.91	7.11	11.2	16.0
	6.0	1.34	1.72	2.58	4.14	6.02	9.89	14.7
	8.0	1.27	1.56	2.23	3.49	5.07	8.50	13.0
	10.0	1.20	1.42	1.95	2.99	4.35	7.54	12.4
Tin	0.5	1.56	2.08	3.09	4.57	6.04	8.64	
	1.0	1.64	2.30	3.74	6.17	8.85	13.7	18.8
	2.0	1.57	2.17	3.53	5.87	8.53	13.6	19.3
	3.0	1.46	1.96	3.13	5.28	7.91	13.3	20.1
	4.0	1.38	1.81	2.82	4.82	7.41	13.2	21.2
	6.0	1.26	1.57	2.37	4.17	6.94	14.8	29.1
	8.0	1.19	1.42	2.05	3.57	6.19	15.1	34.0
	10.0	1.14	1.31	1.79	2.99	5.21	12.5	33.4

Table 2: Energy Deposition Buildup Factor for Isotropic Point Source in Infinite Medium [5].

Material	E_0 , MeV	μr						
		1	2	4	7	10	15	20
Water	0.255	3.03	6.88	21.8	68.1	154	421	906
	0.5	2.46	4.93	13.4	36.1	71.5	163	306
	1.0	2.13	3.71	7.85	16.8	28.2	51.9	83.8
	2.0	1.85	2.82	4.99	8.67	12.9	20.0	28.6
	3.0	1.74	2.52	4.10	6.57	9.12	13.4	17.9
	4.0	1.59	2.18	3.37	5.18	7.01	10.1	13.0
	6.0	1.46	1.89	2.76	3.98	5.18	7.07	8.83
	8.0	1.38	1.74	2.42	3.36	4.28	5.70	7.00
	10.0	1.31	1.60	2.13	2.88	3.60	4.73	5.76
Iron	0.5	2.80	4.84	9.97	20.4	34.2	64.3	101
	1.0	2.19	3.58	7.00	13.6	21.9	38.8	58.5
	2.0	1.78	2.64	4.53	7.90	11.9	19.3	27.4
	3.0	1.58	2.21	3.61	6.05	8.82	14.0	19.8
	4.0	1.45	1.95	3.03	4.90	7.10	11.1	16.7
	6.0	1.30	1.63	2.38	3.73	5.35	8.67	12.7
	8.0	1.21	1.45	2.09	3.39	4.73	6.89	10.4
	10.0	1.16	1.35	1.78	2.61	3.69	6.17	9.89
	10.0	1.16	1.35	1.78	2.61	3.69	6.17	9.89
Lead	0.5	1.51	1.80	2.19	2.61	3.01	3.52	-
	1.0	1.76	2.37	3.39	4.74	6.01	7.78	9.70
	2.0	1.58	2.10	3.14	4.70	6.32	9.05	12.0
	3.0	1.37	1.73	2.50	3.85	5.43	8.65	12.6
	4.0	1.24	1.49	2.09	3.25	4.79	8.46	13.9
	5.11	1.19	1.39	1.89	2.96	4.61	9.38	18.5
	6.0	1.14	1.29	1.70	2.63	4.20	9.46	21.5
	8.0	1.10	1.21	1.50	2.21	3.50	8.60	23.0
	10.0	1.08	1.16	1.38	1.92	2.92	7.15	20.4

Table 3: The mass attenuation coefficient (μ/ρ) for water [12]

Energy (MeV)	Mass attenuation coefficient (μ/ρ) (cm^2/g)	Energy (MeV)	Mass attenuation coefficient (μ/ρ) (cm^2/g)
0.5	0.0966	3.0	0.0396
0.6	0.0896	4.0	0.0339
0.8	0.0786	5.0	0.0301
1.0	0.0706	6.0	0.0275
1.5	0.0575	8.0	0.0240
2.0	0.0493	10.0	0.0219

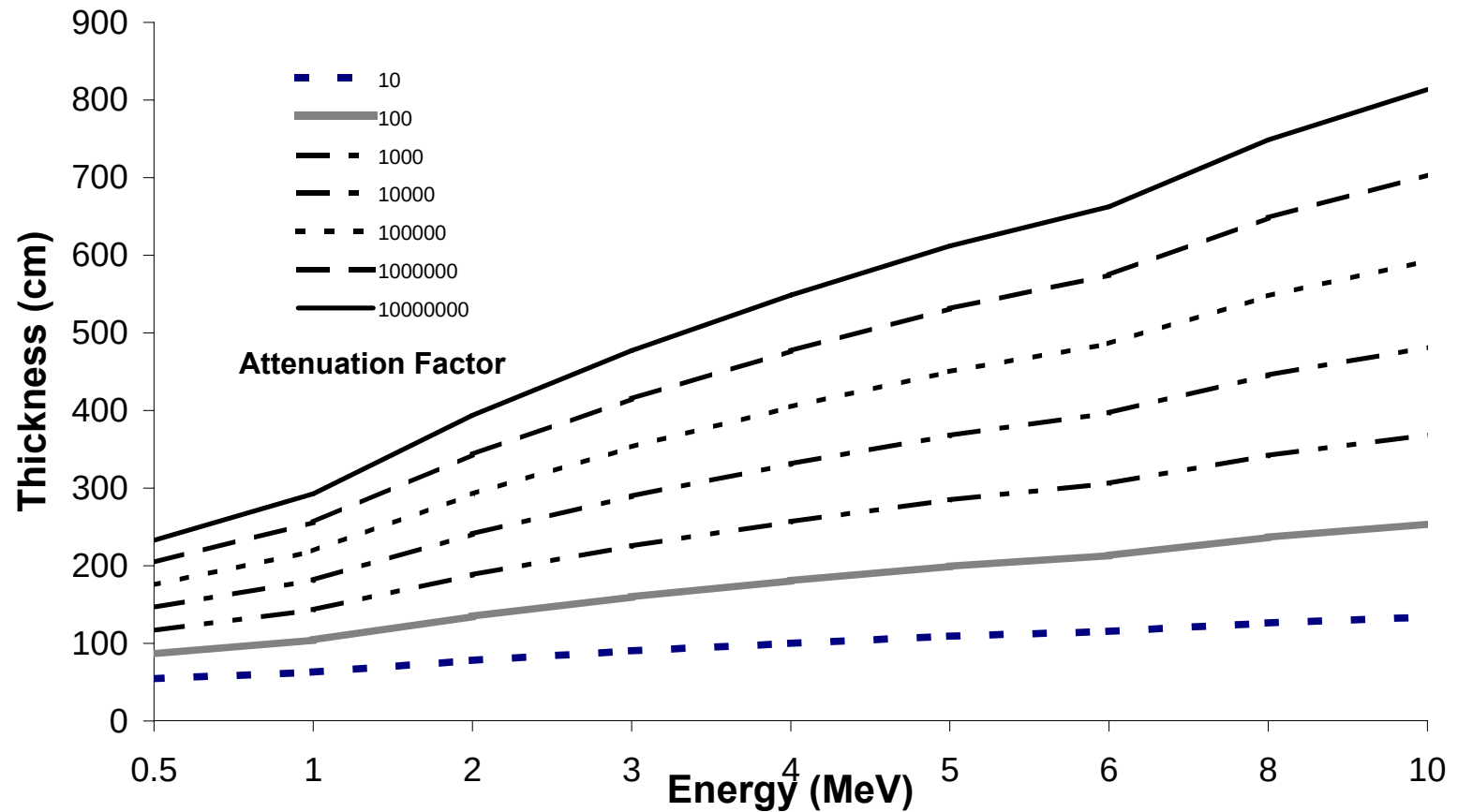
Table 4: Parameters for Taylor form of the exposure build-up factor for point isotropic source for water [5].

Energy (MeV)	A	$-\alpha_1$	α_2
0.5	100.845	0.12687	-0.10925
1.0	19.601	0.09037	-0.02522
2.0	12.612	0.05320	0.01932
3.0	11.110	0.03550	0.03206
4.0	11.163	0.02543	0.03025
6.0	8.385	0.01820	0.04164
8.0	4.635	0.02633	0.07097
10.0	3.545	0.02991	0.08717

Table 5: Calculated thickness (cm) of Water (density=1 g. cm⁻³) for different attenuation factor and γ rays energies with the help of “W-Shielder” program.

Attenuation factor	Energy (MeV)								
	0.5	01	02	03	04	05	06	08	10
10	54.26	62.9	77.84	90.22	99.99	109.2	115.39	126.21	133.99
10²	86.3	104.1	134.7	159.80	180.56	199.29	213.10	236.59	253.78
10³	116.68	143.23	188.51	225.62	256.92	284.80	306.10	342.21	368.86
10⁴	146.26	181.34	240.85	289.71	331.30	368.14	396.81	445.46	481.61
10⁵	175.36	218.83	292.37	352.77	404.49	450.17	486.15	547.34	592.98
10⁶	204.13	255.93	343.34	415.18	476.91	531.36	574.58	648.31	703.48
10⁷	232.67	292.75	393.94	477.12	548.77	611.92	662.37	748.7	813.39

Fig. 14. Water Column Thicknes for Different Gamma Energies and Attenuation Factor



9 REFERENCES:

1. Paul-Friedrich Sauermann, Radiation Protection by Shielding, Nuclear Research Center Julich, Verlag Karl Thiernig, Munchen, Germany, 1976.
2. Jahan Zeb, Shielder, Gamma Shielding calculations of radionuclides emitting photons 0.5 to 10 MeV, OECD Nuclear Energy Agency (NEA) France, **IAEA1391**, <http://www.nea.fr/abs/html/iaea1391>.
3. Jahan Zeb, Waheed Arshed & S. D. Orfi, C-SHIELDER, Gamma shielding calculations of radionuclides emitting photons 0.5 to 10 MeV by different concretes, OECD Nuclear Energy Agency (NEA) France, **IAEA1403**, 01-07-2003. <http://www.nea.fr/abs/html/iaea1403.html>.
4. Jahan Zeb, Waheed Arshed & S. D. Orfi, W-SHIELDER, calculates shielding thickness of water for photon emitting radionuclides between 0.5 to 10 MeV, OECD Nuclear Energy Agency (NEA) France, **IAEA1417**, 03-06-2005. <http://www.nea.fr/abs/html/iaea1417.html>.
5. James Wood, Computational Methods in Reactor Shielding, Pergamon press, U.K. 1st edition, 1982
6. R. D. Evans, "The Atomic Nucleus," p.712, McGraw-Hill Book Company, Inc., New York, 1955.
7. J. R. Harrison, Nuclear Reactor Shielding, Nuclear Engineering Monographs, Temple Press Limited, London, U. K. 1958.
8. Goldstein, H. and J. E. Wilkins (1954). USAEC Report NYO-3075.
9. Trubey, D. K. (1966). A Survey of Empirical Functions used to fit Gamma Ray Buildup Factors. USAEC Report ORNL-RSIC-10.
10. R. G. Jaeger, E. P. Blizard, A.B. Chilton, M. Grotenhuis, A. Honig, Th. A. Jaeger, H. H. Eisenlohr, Engineering Compendium on Radiation Shielding, volume I, Springer-Verlag Berlin Heidelberg, New York, U.S. A. 1968.
11. Schaeffer, N. M. (Ed.) (1973). Reactor Shielding for Nuclear Engineers. USAEC Report TID-25951.
12. Nasir Ahmed, Matiullah, Muhammad Tufail, Mutawarra Hssain, (1990), "Useful Data for Nuclear Engineers", CNS-22, PIEAS.

10. BIBLIOGRAPHY OF W-SHIELDER (SOFTWARE)

Activity: Number of spontaneous nuclear disintegrations occurring in a given quantity of material during a suitably small interval of time divided by that interval of time. The SI derived unit of activity is the Becquerel (Bq). $1 \text{ Bq} = 1 \text{ d/s}$. An old unit of activity is Curie, which is equal to $3.7 \times 10^{10} \text{ Bq}$.

Exposure: For γ radiation in air, the sum of the electrical charges of all of the ions of one sign produced in air when all electrons liberated by photons in a suitably small element of volume of air are completely stopped in air, divided by the mass of the air in the volume element. It is commonly expressed in Roentgens (R).

$$1 \text{ R} = 2.58 \times 10^{-4} \text{ C/kg}$$

Gamma Factor: It is defined as, exposure in Roentgen per hour at the surface of a sphere having radius of 1 cm or 1 m from a source of 1 mCi or 1Ci placed at the center of sphere.

$$\Gamma = R \cdot \text{cm}^2/\text{h.mCi}$$

Or

$$\Gamma = R \cdot \text{m}^2/\text{h.Ci}$$

Reduced Exposure: Exposure on the surface of shielding material (Water) or exposure, which comes out after the source, is shielded.

A: Taylor build-up coefficients.

α_1 : Taylor build-up coefficients

α_2 : Taylor build-up coefficients

μ : linear attenuation coefficient

μ_α : linear absorption coefficient